

1. Let X be a data set such that standard deviation (S.D) of $X=6$. Then, S.D. of $\frac{4X+6}{4}$ is ?
a) 7 **b) 6** c) 0 d) 15
2. The sum of $\sin^2 1^\circ + \sin^2 2^\circ + \dots + \sin^2 89^\circ + \sin^2 90^\circ$ is which of the following?
a) 44 **b) $\frac{91}{2}$** c) 45 d) $\frac{89}{2}$
3. Three points $A(1, -2)$, $B(3,4)$ and $C(4,7)$ forms?
a) Straight line b) Right angled triangle c) An equilateral triangle d) None
4. If $k < 1$, which of the following must be negative?
a) $k^2 - 6k + 8$ b) $k^2 - 2k + 1$ c) $|k| - k^2$ **d) $2^k - 3$**
5. If the arithmetic mean of 3, 8 and w is greater than or equal to w and smaller than or equal to $3w$, how many integer values of w exist?
a) Five **b) four** c) three d) two
6. How many rectangles exist in (8×8) chess board?
a) 1296 b) 64 c) 256 d) 512
7. What is the last digit of $1! + 2! + 3! + 4! + \dots + 100!$?
a) 3 b) 2 c) 1 d) 4
8. The ratio of the marbles with Jack and Dave was 13:9. Now, they each have 10 more marbles with them. Which of the following cannot be the ratio of their marbles now?
a) 7:4 b) 7:5 c) 15:11 d) 31:23
9. If $\frac{1}{2} f(x) = f\left(\frac{1}{2}x\right)$, which of the following is true for all values of $f(x)$?
a) $f(x) = 2x + 2$ b) $f(x) = x^2$ c) $f(x) = \sqrt{x}$ **d) $f(x) = 13x$**
10. A total of \$450 was donated to charity by 25 employees. If 15 employees donated at least \$12 but less than \$19 and 9 employees donated at least \$19, what is the maximum amount, in dollars, that the last employees could have donated?
a) \$51 **b) \$99** c) \$109 d) \$89
11. The price of a phone call consists of a standard connection fee, which is constant, plus a per-minute charge. A ten minute call costs Rs 2.90 and a sixteen minute call costs Rs 4.40. How much does a thirteen minute call cost?
a) Rs 3.55 b) Rs 3.57 c) Rs 3.58 **d) Rs 3.65**
12. $\frac{4}{5}$ Of the women and $\frac{3}{4}$ of the men in a group speak Spanish. If there are 40% as many men as women in the group, what fraction of the group speaks Spanish?
a) $\frac{14}{11}$ **b) $\frac{11}{14}$** c) $\frac{3}{5}$ d) $\frac{7}{11}$
13. Which one of the following sets is equal to $\left[\frac{1}{2}, 1\right) \cup \left[\frac{1}{3}, 1\right) \cup \left[\frac{1}{4}, 1\right) \cup \dots$?
a) (0, 1) b) $[0, 1)$ c) $(0, 1]$ d) $[0, 1]$
14. How many factors greater than 1 do 120, 210 and 270 have in common?
a) 1 b) 3 c) 6 **d) 7**

15. Ramon wants to cut a rectangular board into identical square pieces. If the board is 18 inches by 30 inches, what is the least number of square pieces he can cut without wasting any of the board?
 a) 6 b) 9 c) 12 **d) 15**
16. If $125^{14}48^8$ were expressed as an integer, how many consecutive zeros would that integer have immediately to the left of its decimal point?
 a) 22 **b) 32** c) 42 d) 50
17. The area of the region bonded by the curve: $8x^2+6y^2-16x+12y+13=0$ is?
 a) $\frac{\pi}{\sqrt{40}}$ sq. unit b) $\frac{\pi}{\sqrt{60}}$ sq. unit c) $\frac{\pi}{\sqrt{36}}$ sq. unit **d) $\frac{\pi}{\sqrt{48}}$ sq. unit**
18. How many whole numbers are between $\sqrt{77}$ and $\sqrt{257}$?
 a) 7 b) 10 c) 9 **d) 8**
19. If the price of petrol is increased by 10% and then decreased by 10%, what is the equivalent change in percentage?
 a) 0% b) 1% increased **c) 1% decreased** d) None of them
20. The domain of the function defined by $f(x) = \sqrt{x-2}$ is
 a) $\mathbb{R}$ **b) $[2, \infty)$** c) $(2, \infty)$ d) $\mathbb{R}-\{3\}$
21. If a, b, c are in AP, a, x, b and b, y, c are in GP then x^2, b^2, y^2 are in
a) AP b) GP c) HP d) None
22. In a parallelogram ABCD, OA and OB are the bisectors of the angles A and B respectively at O, then an angle AOB equals to ?
 a) 60° b) 75° **c) 90°** d) 120°
23. For which of the following positive integers is the square of that integer divided by the cube root of the same integer equal to nine times that an integer?
 a) 4 b) 8 c) 16 **d) 27**
24. If $|x+1| = 2|x-1|$, what is the value of $|x - \frac{5}{3}|$?
 a) $\frac{1}{3}$ b) 3 **c) $\frac{4}{3}$** d) $\frac{7}{3}$
25. If $4x < x < x^3 < x^2$, which of the following is possible value of x ?
 a) $-\sqrt{3}$ b) $-\frac{3}{2}$ c) -1 **d) $-\frac{4}{5}$**
26. In the XY-plane, what is the area of the triangle formed by the line $3y - 4x = 24$ and the X and Y axes?
 a) 6 b) 14 **c) 24** d) 36
27. A bag contains an equal number of 5 rupee, 1 rupee, 50 paisa & 25 paisa coins respectively. If the total value is 135, how many coin of each type of there?
 a) 22 b) 19 c) 21 **d) 20**
28. Auto lot has a 2:1 ratio of blue cars to red cars and a 6:1 ratio of red cars to orange cars on the lot. What could be the total number numbers of blue, red and orange cars on the lot?
 a) 37 **b) 38** c) 39 d) 40
29. Each of 5 people had a blank card on which they wrote a positive integer. If the average (Arithmetic mean) of these integers is 15, what is the greatest possible that could be on one of the cards?
 a) 70 **b) 71** c) 72 d) 65

30. Which of the following choices of 'a' will results in a system of equations $9x - 14y = -3$, $2x - ay = -6$ with no solution?
 a) $-\frac{9}{14}$ b) $-\frac{28}{9}$ c) $\frac{9}{14}$ **d) $\frac{28}{9}$**
31. The original price of an article was reduced by 20%. During a special sale, the new price was decreased by 10%. The price of an article now has to bring back to the original price by 2 successive mark ups. If the second mark up is 25%, what is the first mark up approximately?
 a) 5% b) 16.67% **c) 11.11%** d) 12.5%
32. If 8 points (A, B, C, D, E, F, G, H) lies on the line L and 4 points (P, Q, R, S) lies on the line K. Line L and line K are parallel. How many different possible triangles can be connecting any three points?
 a) 155 b) 158 **c) 160** d) 162
33. What is the minimum value of the function $y = x^2 + 6x + 13$ in the real domain?
 a) 3 b) 5 c) 6 **d) 4**
34. If P is a perimeter of a triangle with one side of 7 and another side of 9, what is the range of possible values for P?
 a) $2 < P < 16$ b) $3 < P < 17$ **c) $18 < P < 32$** d) $18 < P < 33$
35. A sector of a circle has a radius of 10 and an area 20π . What is the arc length of a sector?
 a) π b) 2π **c) 4π** d) 5π
36. How many liters of pure Alcohol must be added to a 90 liters solution that is 20% Alcohol in order to produce a solution that is at least 25% Alcohol?
 a) 4.5 b) 5 c) 5.5 **d) 6.5**
37. A badminton club has 21 members. What is the ratio of number of 6 member committees that can be formed from the members of the club to the number of 5 member committees that can be formed from the members of club?
 a) 17 to 16 b) 15 to 1 **c) 8 to 3** d) 17 to 6
38. Kabir's college is 12 miles from his hostel. He travels 6 miles from the college to basketball practice, and then 4 miles to a computer class. If he is D miles from home, what is the range of possible values for D?
 a) $12 \leq D \leq 10$ b) $2 \leq D \leq 16$ c) $10 \leq D \leq 22$ **d) $2 \leq D \leq 22$**
39. Which of the following gives the correct values of x, for which the expression $\sqrt{\frac{2}{(x-1)(x+2)}}$ is defined?
 a) $-2 \leq x \leq 1$ b) $x \geq 1$ **c) $x > 1$** d) $x \leq -2$
40. Seven consecutive numbers are selected from the integers 1 to 100, and each number is divided by 7. What is the sum of remainder obtained?
a) 21 b) 20 c) 22 d) 19
41. ABCD is a rectangle. Points P, Q, R, S and T cut side AB of the rectangle such that AD=9, AB=15, AP=3, PQ=QR=RS=ST=1. E is a point on AD such that AE=3. Which of the following segments is parallel to the diagonal BD of the rectangle?
 a) EQ **b) ER** c) ES d) ET

42. Running on a 10-mile loop in a same direction, Sue ran at a constant rate of 8 miles per hour and Rob ran at a constant rate of 6 miles per hour. If they began running at the same point on the loop. How many hours later did sue complete exactly 1 more lap than Rob?
- a) 3 b) 4 **c) 5** d) 6
43. The center of a circle is (10, -3). The point (10, 9) is outside the circle, and the point (6, -3) is inside the circle; neither point is on the circle. If the radius r , is an integer, how many possible values are there for r ?
- a) Seven** b) Eight c) Nine d) Ten
44. When three positive integers a , b , c are multiplied together, their product is 100. Suppose $a < b < c$. In how many ways the number be chosen?
- a) 2 b) 3 **c) 4** d) 5
45. The traffic lights at three different road crossing changes after 48sec, 72sec & 108sec. If they all changes simultaneously at 8:20:00 hours, at what time will they again changes simultaneously?
- a) 8:30:12 hrs. b) 8:29:12 hrs. c) 8:28:12 hrs. **d) 8:27:12 hrs.**
46. If an ant climbing a pole completes climbing in 5 days. Ant climbs 4 meters in a day and slips down 3 meters in a night. What is the height of pole ?
- a) 9m **b) 8m** c) 7m d) 5m
47. A rectangular floor having perimeter 16 meters is to be covered with square carpets that measure 1 meter by 1 meter each and cost Rs 6 apiece. What is the maximum possible cost for the number of square carpets needed to cover the rectangular floor if the sides of the floor are integers?
- a) Rs 42 b) Rs 72 c) Rs 90 **d) Rs 96**
48. What is the difference between maximum and minimum value of $\frac{x}{y}$ for which $(x - 2)^2 = 9$ and $(y - 3)^2 = 25$?
- a) $-\frac{15}{8}$ b) $\frac{19}{8}$ **c) $\frac{25}{8}$** d) $\frac{9}{8}$
49. The equation of the image of the line $2x + 3y = 6$ under the reflection by y-axis is
- a) $2x - 3y = 6$ **b) $2x - 3y + 6 = 0$** c) $2x + 3y + 6 = 0$ d) $2x + 3y = 6$
50. If $x^2 - \frac{1}{x^2} = 5 - \sqrt{3}$, then what is the value of $x^2 + \frac{1}{x^2}$?
- a) $5 + \sqrt{3}$ b) $7 + \sqrt{3}$ c) $\sqrt{5 + \sqrt{3}}$ **d) $\sqrt{32 - 10\sqrt{3}}$**